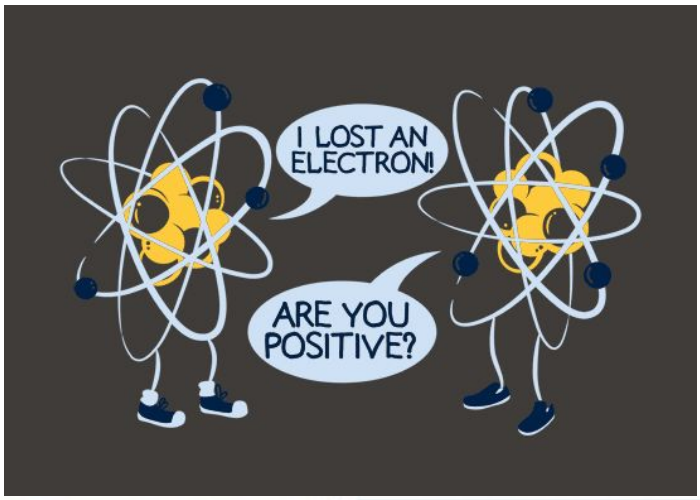
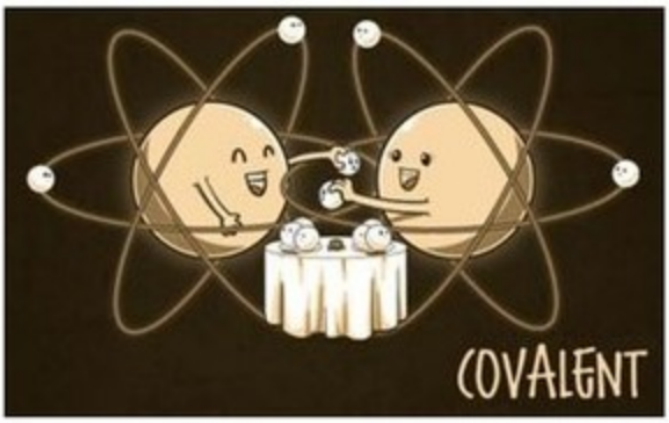
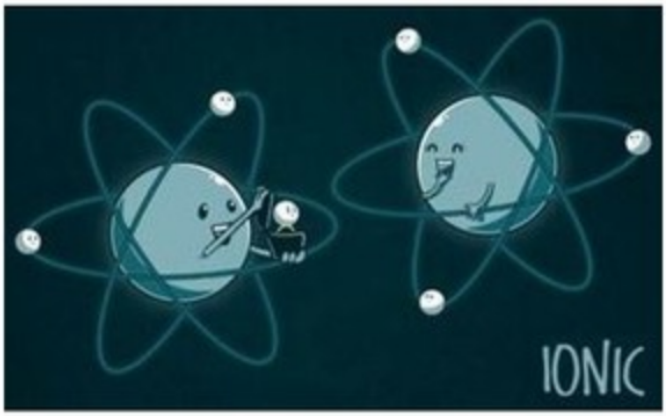
Science 9

Unit 2: Chemistry *(Part 2)*





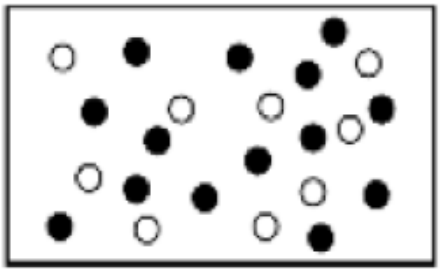


Book 1: Ions & Chemical Bonding

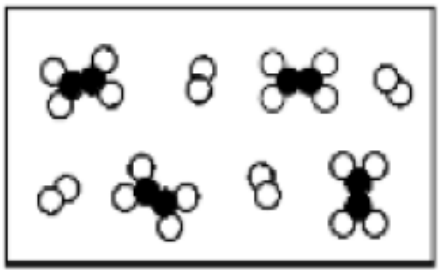
name: _____ block: _____

PART A: FORMING COMPOUNDS

RECALL that matter can be classified as a _____ or a _____.
Pure _____: A type of matter that contains only _____ type of _____.
This particle can be an _____ or a molecule (_____).

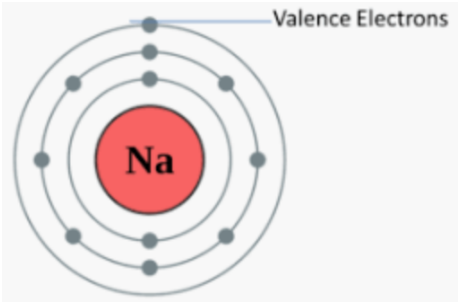


- **Element**: A type of matter that contains only one type of _____.
    - *Examples*:
      _____
      _____
- **Compound**: A type of matter that contains _____ or more _____. These elements are in definite _____ (chemically bonded together).
    - *Examples*:
      _____
      _____



RECALL that _____ are the electrons in an atoms outermost shell, these are the only electrons involved in c_____ b_____.

When two atoms move close together, their valence electrons interact.



A c_____b_____ forms between the atoms if the new arrangement of atoms and electrons is stable.
The stability of an atom, ion, or compound is related to its e_____; lower energy states are more stable.
The ***lowest energy*** is achieved when the atoms in the compound have the same arrangement of valence electrons as a n_____g_____ (full electron shell) to which they are closest in the periodic table.

When an atom forms a compound, forms a valence shell like its closest noble gas in one of *three ways:*

1. Atoms of metals may _____to other atoms, forming c_____
2. Atoms of non-metals may _____ electrons from other atoms, forming a_____
3. Atoms may _____electrons. *(covalent bonding)*



PART B: WHAT IS AN ION?

RECALL that **ATOMS** are _____, meaning they have no charge. The number of protons (_____ charge) is equal to the number of electrons (_____ charge)

atomic charge = number of protons - number of electrons

ATOMIC CHARGE

- The overall charge of an ***atom*** is _____ **(the # of _____ = the # of _____)**
- The charge of an atom can be _____
  - If electrons are _____, there is a negative charge
  - If electrons are _____, there is a positive charge
  - When there is a charge the atom is called an _____
    If the overall charge of an *ion* is _____, then you have more _____ than _____
    If the overall charge of an *ion* is _____, then you have more _____ than _____
- _____ are NEVER added or removed from an atom in a chemical reaction!!!



Symbol	Atom or Ion?	# of Protons	# of Electrons	Atomic Charge
Li	Atom			
Li^{+1}				
Cr^{+3}				
I^{-1}				
Ar				
			36	0
		64	64	
		55	54	
P^{-3}				-3

PRACTICE

Problem: If an ion of sulfur has a charge of 2-, how many electrons does it have?

Ion Charge = Number of _____ – Number of _____

PART C: ION FORMATION

When atoms_____ or lose _____, they become electrically charged particles called _____.

Metal atoms, for example, _____electrons to form _____charged ions called _____.

Many metals can form a cation only in one way.

For example, aluminum forms a cation by losing three electrons to become _____.

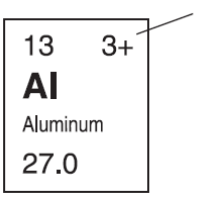
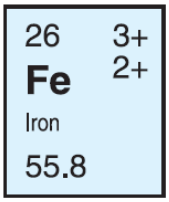


Figure 4.4A Aluminum forms a 3+ ion as a result of losing three electrons.

Some metals are _____, which means they can form ions in ***more than one way***, depending on the chemical reaction they undergo.

For example, iron is a ***multivalent*** element because it can lose _____ or t_____ electrons to become _____ ions and_____ ions.



Look at the periodic table. Which metals are multivalent?

_____

_____ Many _____ also form ions.

However, since non-metal atoms, with very few exceptions, _____electrons, they form _____ions called _____.

For example, the periodic table shows that chlorine will form a _____ion. This happens when a chlorine atom _____one electron.

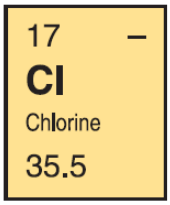


Figure 4.4C Chlorine forms a negative ion.

PRACTICE

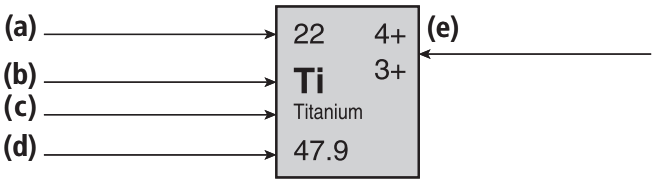
Review....AND NEW

What is in the box?

Test your knowledge how information is displayed for each element in the periodic table.

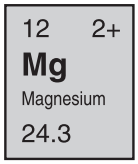
1. Use the vocabulary words listed to label the diagram.

Vocabulary	
ion charge	name
atomic number	symbol
average atomic mass	



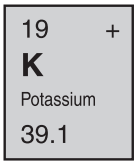
Examine the periodic table entry for each of the following elements and complete the blanks below.

2.



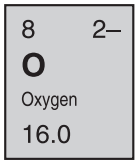
(a) atomic number _____
(b) average atomic mass _____
(c) ion charge _____
(d) number of protons _____

3.



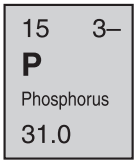
(a) name of element _____
(b) ion charge _____
(c) number of protons _____
(d) average atomic mass _____

4.



(a) atomic number _____
(b) average atomic mass _____
(c) ion charge _____
(d) symbol of element _____

5.



(a) name of element _____
(b) average atomic mass _____
(c) ion charge _____
(d) number of protons _____

PART D: ION PATTERNS IN THE PERIODIC TABLE

The outermost shell that contains electrons is called the _____. The electrons in the valence shell are called the _____electrons.

Valence electrons are involved in c_____b_____.

FOR EXAMPLE...

- The atoms of each element in group 1, hydrogen, have only _____ electron in their valence shell.
- Group 2 elements have _____electrons in their valence shell.
- Group 13 elements have _____ electrons in their valence shell.
- Group 14 has _____electrons, and so on through group 18.
- All group 18 elements have filled valence shells. Helium has two electrons filling its valence shell. Neon and argon each have eight electrons, or a stable octet, filling their valence shell.

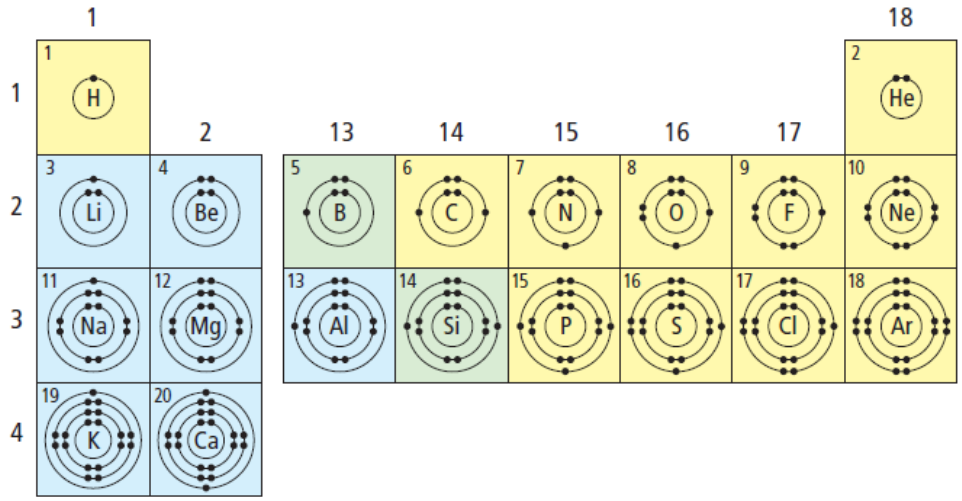


Figure 4.7 Occupied shells for individual atoms of the first 20 elements in the periodic table. Note that the diagrams do not represent the position or path of electrons.



Element	Atomic Number	Ion Charge	Atom or Ion	Number of Protons	Number of Electrons
Beryllium	4	2+	Ion	4	2
	11	0	Atom		
		0		18	
Chlorine		0			
		3-			10



ASSIGNMENT #1: Atomic & Ions Practice
This assignment is to be completed below in the space provided.

Remember your formulas:

Formula 1. atomic number = number of protons

Formula 2. atomic mass = number of protons + number of neutrons
number of protons = atomic mass – number of neutrons
number of neutrons = atomic mass – number of protons

Formula 3. atomic charge = number of protons – number of electrons
number of protons = number of electrons + atomic charge
number of electrons = number of protons – atomic charge

Element Name	Atomic Number	Atomic Mass	Number of Protons	Number of Electrons	Number of Neutrons	Atomic Charge
beryllium	4	9	4	4	5	0
	11		11			0
				18		0
calcium						0
chlorine						0
	47					0
		197				0
beryllium	4	9	4	2	5	+2
	7			10		-3
	16					-2
	3		3			+1
			13			+3
nitrogen				10		
		16		10		
			15	18		
	13			10		

PART E: ATOMS VS IONS

Remember…
Atoms are NEUTRAL with equal _____ = _____ ,whereas IONS have a **charge** due to_____ or _____ electrons.

Lithium **Atom**	Lithium **Ion**
Oxygen **Atom**	Oxygen **Ion**

NAMING IONS

1) Naming **METAL IONS**
   - **use the name of the metal, and add the word "ion" at the end.**

Example: Sodium metal (Na) forms the _____
Aluminum metal (Al) forms the _____

1) Naming **NON-METAL IONS**
   - **take off the original ending of the name, and add "_____"**
   -

Element name	Element symbol	Ion name	Ion symbol
fluor**ine**	F	fluor	
chlor**ine**	Cl	chlor	
brom**ine**	Br	brom	
iod**ine**	I	iod	
ox**ygen**	O	ox	
sulph**ur**	S	sulph	
nitr**ogen**	N	nitr	
phosph**orus**	P	phosph	

3) Naming **MULTIVALENT METAL IONS**

- **Multivalent Ions:** most ***transition metals*** are multivalent, meaning they have more than 1 stable state.
- if a metal has more than one possible charge, the charge is shown by using a **R**_____ **N**_____ *after* the name of the metal, and *before* the word "ion"

Example: Fe^{3+} = iron_____, Fe^{2+} = iron_____, U^{6+} = uranium_____, U^{3+} = uranium_____

PRACTICE

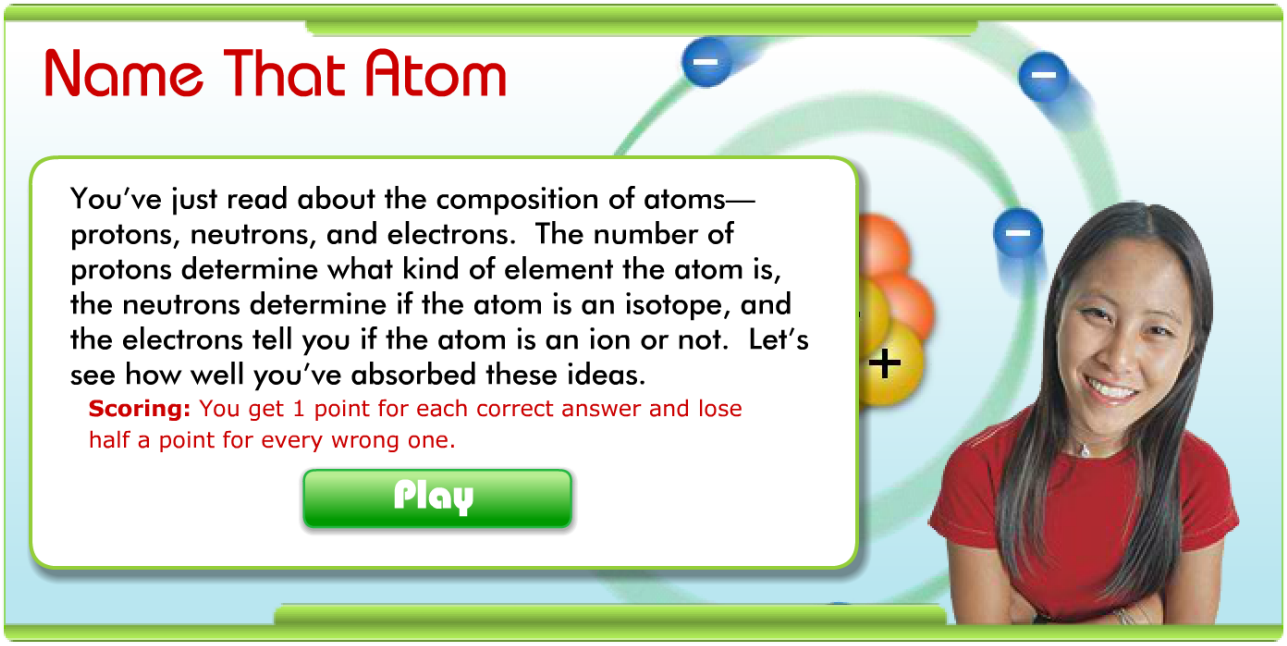
2. Write the names of the following ions using the Stock system of notation.
   (a) Cu^{+} (b) Cr^{3+} (c) W^{6+}

3. Write the formula of the following ions to show their charges.
   (a) cobalt(III) ion (b) nickel(II) ion (c) vanadium(V) ion

Homework

ASSIGNMENT #2:"Name That Atom" ***(link on class website)*** **+ Bohr Diagrams Atoms to Ions pg 10-11**
This assignment is to be completed below in the space provided.

http://www.learner.org/interactives/periodic/basics_interactive.html



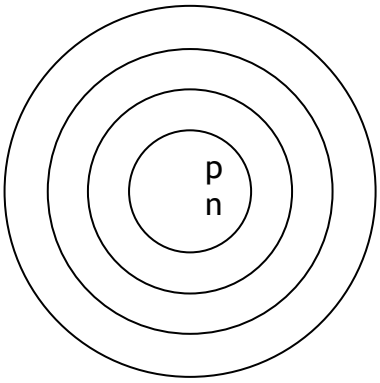
Homework Bohr Diagrams :ATOMS to IONS

For each element:

1. Write the ***standard atomic notation*** in the box.
2. Find the number of protons, neutrons, and electrons in each atom.
3. Draw the Bohr diagram.
4. Find the ion that this atom would create: name it & write it in the box.
5. Draw the diagram for the ion created from the element.

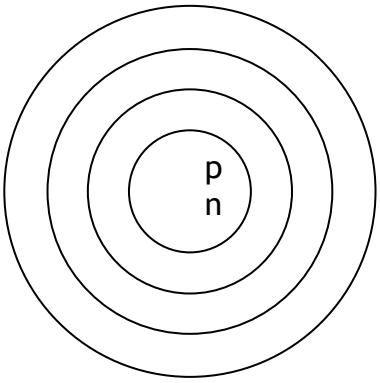
Sodium

p+
n
e-



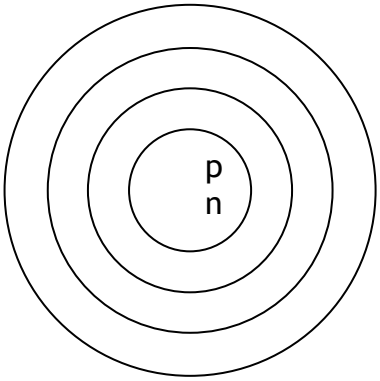
_____ION

p+
n
e-



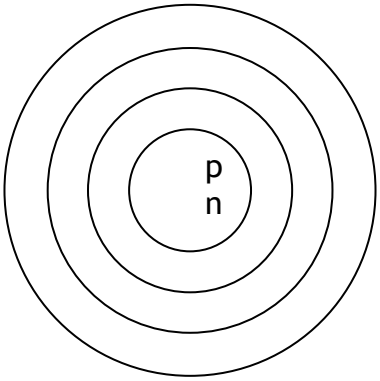
Beryllium

p+
n
e-



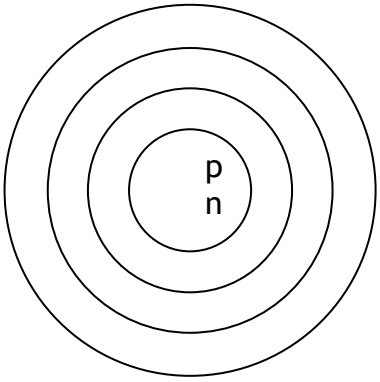
_____ION

p+
n
e-



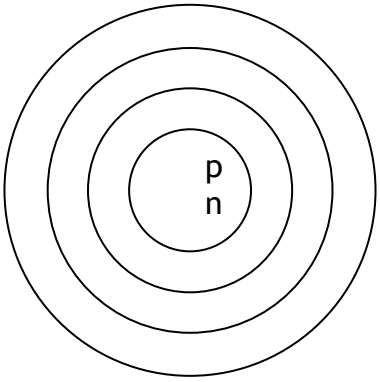
Magnesium

p+
n
e-

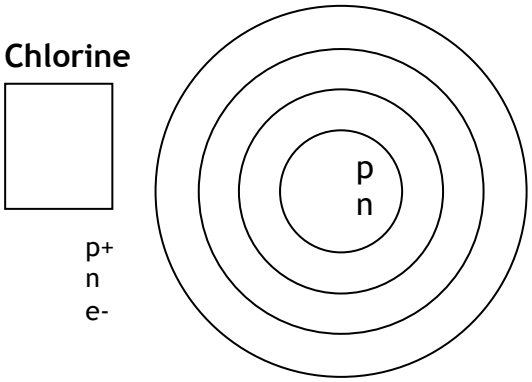


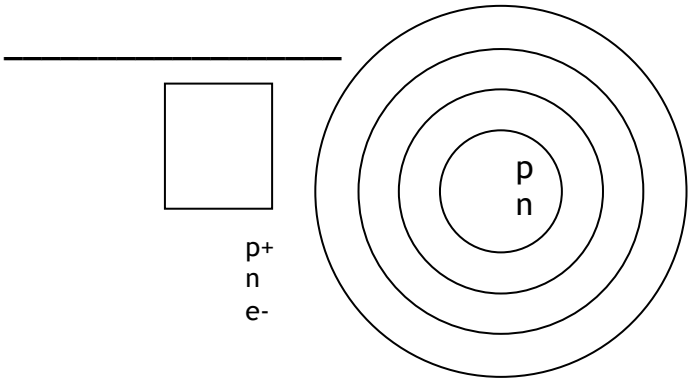
_____ION

p+
n
e-

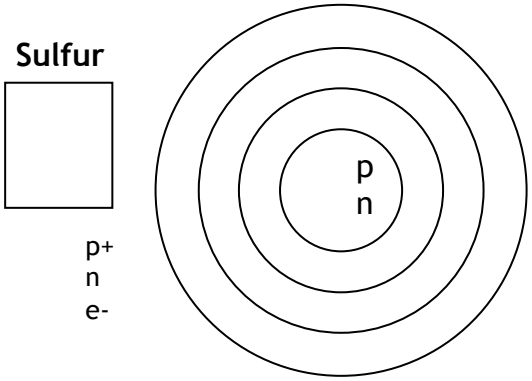


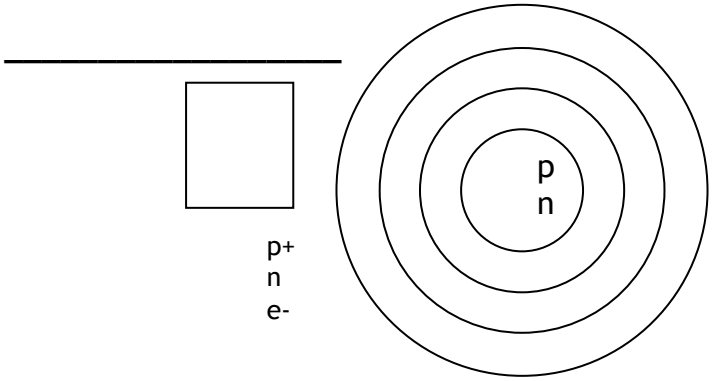
Chlorine



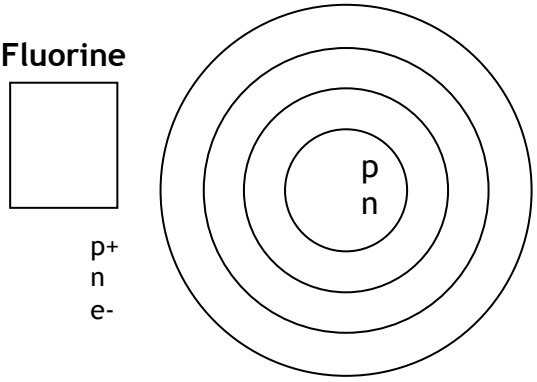


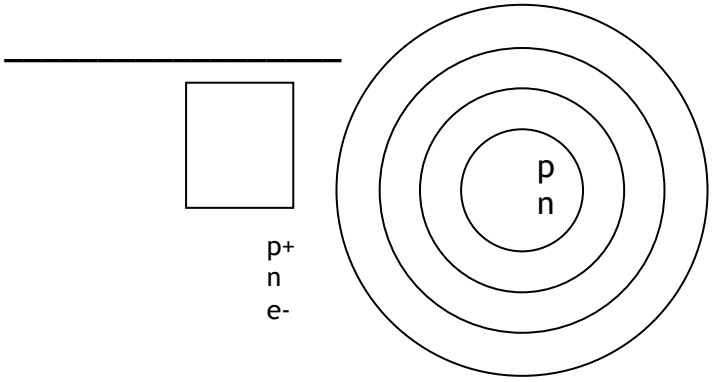
Sulfur





Fluorine





PART F: POLYATOMIC IONS

nitrite NO_2^-
sulphite SO_3^{2-}

nitrate NO_3^-
sulphate SO_4^{2-}

A ***molecule*** is a **neutral** group of *covalently* bonded atoms *(share electrons)*

A **polyatomic ion** is a _____group of covalently bonded atoms so it's like a _____except that it has a _____.

They are relatively stable species that remain intact in chemical reactions. This means the "polyatomic group" does not separate in a chemical reaction when a bond forms.

Many polyatomic ions are _____, consisting of an atom of a given element and some number of _____atoms.

***NOTE* The names & formulae of these are provided in your DATA BOOKLET.**



Polyatomic Practice

1. Name or write the formula for the following polyatomic ions.

sulfate	_____	CO_3^{2-}	_____
nitrite	_____	MnO_3^-	_____
perphosphate	_____	SO_5^{2-}	_____
hypoiodite	_____	BrO_2^-	_____
chlorite	_____	CO_4^{2-}	_____
phosphite	_____	PO_5^{3-}	_____
percarbonate	_____	ClO^-	_____
bromate	_____	IO_2^-	_____
hyposulfite	_____	PO_4^{3-}	_____
permanganate	_____	NO_2^-	_____
carbonite	_____	SO_4^{2-}	_____

Homework

ASSIGNMENT #3: Polyatomic Ions pg 13-15
This assignment is to be completed below in the space provided.

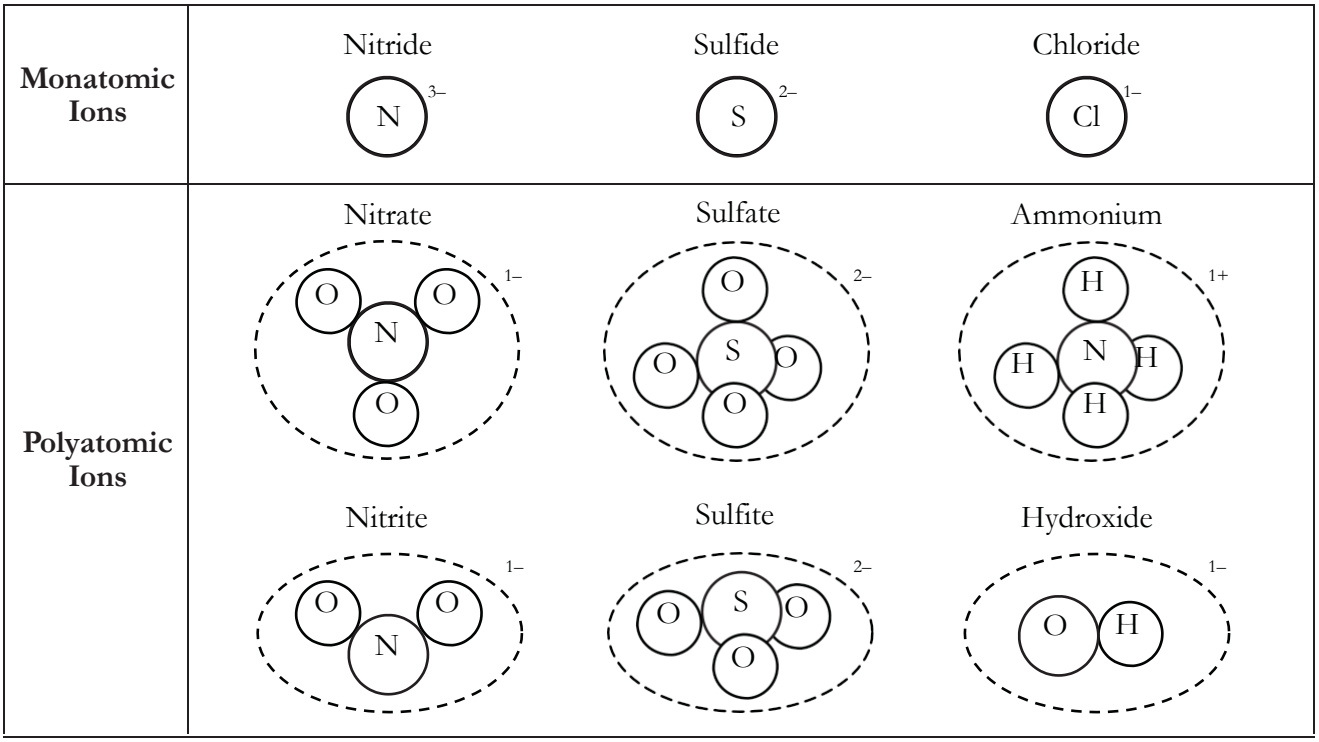
Polyatomic Ions

Why?

Can a group of atoms have a charge?

Do you know you eat a lot of "-ates"? Next time you look at a food label, read the ingredients and you will likely find a number of ingredients that end with "-ate," such as sodium phosphate or calcium carbonate. Did you ever wonder what the chemical formulas of these ingredients look like? In this activity we will explore polyatomic ions, which are groups of atoms that carry a charge. These ions are found in our food ingredients, natural waterways, and many other chemical compounds you encounter every day.

Model 1 – Types of Ions



1. Use Model 1 to complete the table below.

Name of Ion	Nitride	Nitrate	Sulfate	Sulfite	Ammonium
Charge on Ion		-1			
Type and Number of Atoms			1 sulfur 4 oxygen		
Chemical Formula				SO_3^{2-}	

2. Consider the terms "monatomic" and "polyatomic" as they are used in Model 1. Write a definition for each of these terms. It may be helpful to break the words apart (*i.e.,* poly – atomic). Make sure your group comes to consensus.

   Monatomic—

   Polyatomic—

3. What types of elements (metals or nonmetals) are shown in the polyatomic ions in Model 1?

4. What type of bonds (ionic or covalent) hold the atoms together in polyatomic ions? Explain your reasoning.

5. The net charge on a sulfide ion (S^{2-}) is –2. Explain how this ion obtains its charge. Your answer should include a discussion of subatomic particles.



6. The dotted line around each polyatomic ion in Model 1 shows that the group of atoms has a charge. The charge is not on any one atom, but rather on the group of atoms as a whole. Based on your knowledge of monatomic ions, propose an explanation for the net charge on a polyatomic ion. Your answer should include a discussion of subatomic particles.

7. What are the similarities and differences between the nitrate and nitrite ions in Model 1?

8. What are the similarities and differences between the sulfate and sulfite ions in Model 1?

9. The "chlorate" polyatomic ion has a charge of –1 and is composed of one chlorine atom (the central atom) and three oxygen atoms.

   a. Draw a model of a chlorate ion.

   b. Write the chemical formula for the chlorate ion, including its charge.

10. In your group discuss what "chlorite" would look like.

a. Draw a model of a chlorite ion.

b. Write the chemical formula for the chlorite ion, including its charge.

STOP

Model 2 – Common Polyatomic Ions

1+	1–	2–	3–
ammonium NH_4^{1+}	acetate CH_3COO^{1-}	sulfate SO_4^{2-}	phosphate PO_4^{3-}
	hydroxide OH^{1-}	sulfite SO_3^{2-}	
	nitrate NO_3^{1-}	carbonate CO_3^{2-}	
	nitrite NO_2^{1-}	chromate CrO_4^{2-}	
	bicarbonate HCO_3^{1-}	dichromate $Cr_2O_7^{2-}$	
	permanganate MnO_4^{1-}		
	perchlorate ClO_4^{1-}		
	chlorate ClO_3^{1-}		
	chlorite ClO_2^{1-}		
	hypochlorite ClO^{1-}		

11. What is the only polyatomic ion that is a cation?

12. How are bicarbonate and carbonate related?

13. Predict the chemical formula and charge for the bisulfate ion.

14. How are chromate and dichromate related?

15. Bromine forms polyatomic ions with structures similar to those of chlorine. Using the chlorine family of polyatomic ions as a model, predict the name of the BrO_4^{1-} ion.

16. Identify the polyatomic ion in each of these ionic compounds. Write out the name and formula of the ions including their charges.

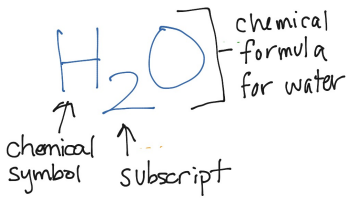
a. $CaCO_3$ *b.* $Mg(OH)_2$ *c.* NH_4Cl

PART 9: MOLECULES & CHEMICAL FORMULA:

- Molecules are particles made up from _____
  - Examples: _____
- Atoms in compounds are held together by _____: _____ between the atoms
  - They can be the same kind of atoms: _____
  - They can be different kinds of atoms: _____
- There are two different kinds of compounds in this unit: _____ & _____

Counting Atoms in Molecules

- When writing chemical formulas, we use: _____ & _____
- **Subscript:**
  - A subscript is a number that is dropped down after a _____.
  - They tell you how many of that _____ are in the compound.
  - Examples: H_2 $C_6H_{12}O_6$



_____ atoms of hydrogen

_____ atoms of carbon

_____ atoms of hydrogen

_____ atoms of oxygen

_____ atoms in total

- **Coefficient:**
  - A coefficient is a big number _____ of an atom or compound.
  - It tells you how many of that _____ is involved in a chemical reaction.
  - Example: $6CO_2$ Coefficient: _____ Subscript: _____

    In one molecule, there are _____ atoms of carbon

    In one molecule, there are _____ atoms of oxygen

    In total, there are _____ atoms of carbon

    In total, there are _____ atoms of oxygen

  - Example: $4C_3H_8$

    In total, there are _____ atoms of carbon

    In total, there are _____ atoms of hydrogen

Using Brackets

- Brackets separate out smaller _____ within a larger _____ thereby allowing subscripts to be used just within the bracketed area.
    - Example: $Sr(NO_3)_2$
        - In words: $Sr(NO_3)_2$ is the equivalent of: Three _____ atoms attached to each _____ atom and two of these structures are attached to one strontium ion.

Calculate the number of atoms of each element in ONE molecule and in FIVE molecules of $Sr(NO_3)_2$ (strontium nitrate).

One molecule		**Five molecules**	
_____	Strontium	Sr	5 x _____ = _____
_____	Nitrogen	N	5 x _____ = _____
_____	Oxygen	O	5 x _____ = _____
Total = _____		Total = _____	



Count the total number of atoms in each of the following molecules/ions.

1. BaS _____
2. N_2O_4 _____
3. CN^- _____
4. ClO_2^- _____
5. $NaNO_3$ _____
6. CH_4 _____
7. $SnCl_2$ _____
8. Na_2Se _____
9. NH_3 _____
10. NO_3^- _____
11. PO_4^{3-} _____
12. OCl_2 _____
13. PbO_2 _____
14. $FeCl_3$ _____
15. SO_2 _____
16. S_2F_{10} _____
17. OH^- _____
18. K_3P _____
19. $(NH_4)_2S$ _____
20. NO_3 _____
21. NO _____
22. NH_4^+ _____
23. $BeBr_2$ _____
24. PdI_4 _____



ASSIGNMENT #4: Counting Atoms in Molecules
This assignment is to be completed below in the space provided.

Complete the chart below by filling in the missing information. The first formula has been completed as an example.

Formula	Number of elements	Names of the elements	Number of atoms of each element	Total number of atoms in molecule
MgO	2	magnesium oxygen	1 1	2
SO_2				
NH_3				
H_2O_2				
Fe_2O_3				
H_2CO_3				
$(NH_4)_3P$				
$Ca(OH)_2$				
$MgSO_4$				

PART H: TYPES OF COMPOUNDS

FORMING COMPOUNDS

When two atoms move close together, their valence electrons interact.

A c_____ b_____ forms between the atoms if the new arrangement of atoms and electrons is stable.

A compound forms when electrons are _____ or "_____" between two or more different elements in regular, repeating proportions.

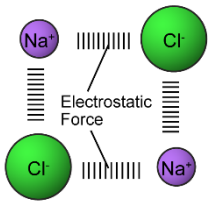
When an atom forms a compound, it may acquire a valence shell like its closest noble gas in one of three ways:

1. Atoms of metals may _____to other atoms, forming c_____
2. Atoms of non-metals may _____ electrons from other atoms, forming a_____
3. Atoms may _____electrons. *(covalent bonding)*

TYPES OF BONDS

Ionic Compounds

- Ionic compounds contain only _____ parts – a _____ and a _____.
- Ionic compounds form when electrons are "stolen" or _____.
- An electrostatic _____ holds ionic compounds together.



Covalent Compounds

- Covalent compounds contain only _____ parts – a _____ and a _____.
- Covalent compounds _____ electrons.

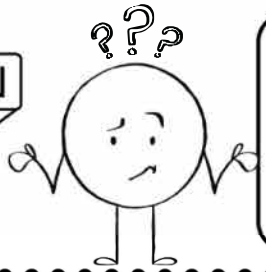


Ionic or Covalent Compound?

For each of the following questions, determine whether the compound is ionic or covalent and write an I (ionic) or C (covalent) on the line beside.

1)	Na_2CO_3	_____	6)	$GaCl_3$	_____
2)	P_2O_5	_____	7)	$CoBr_2$	_____
3)	NH_3	_____	8)	B_2H_4	_____
4)	$FeSO_4$	_____	9)	CO	_____
5)	SiO_2	_____	10)	P_4	_____

CHEMICAL BONDING



WHY DO BONDS FORM?

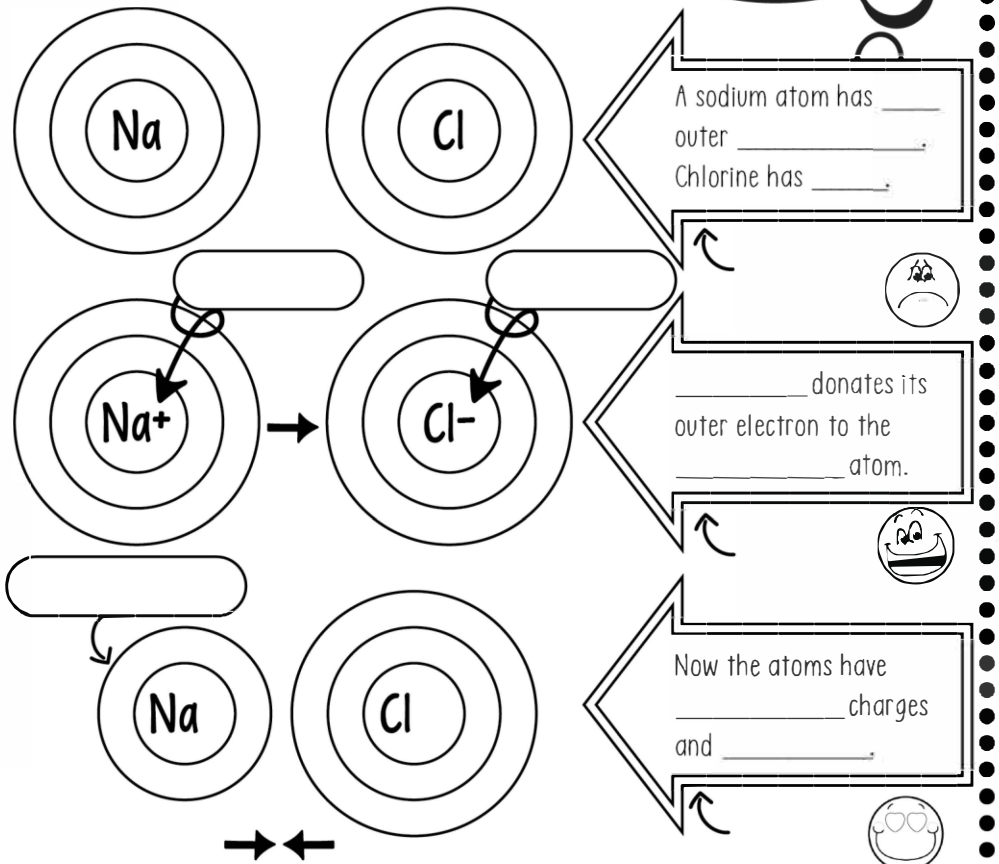
Atoms bond because they become more _____ when they have _____ outer _____. Donating, _____, or sharing _____ allows them to achieve _____. The _____ Rule - 8 electrons in the _____ shell.

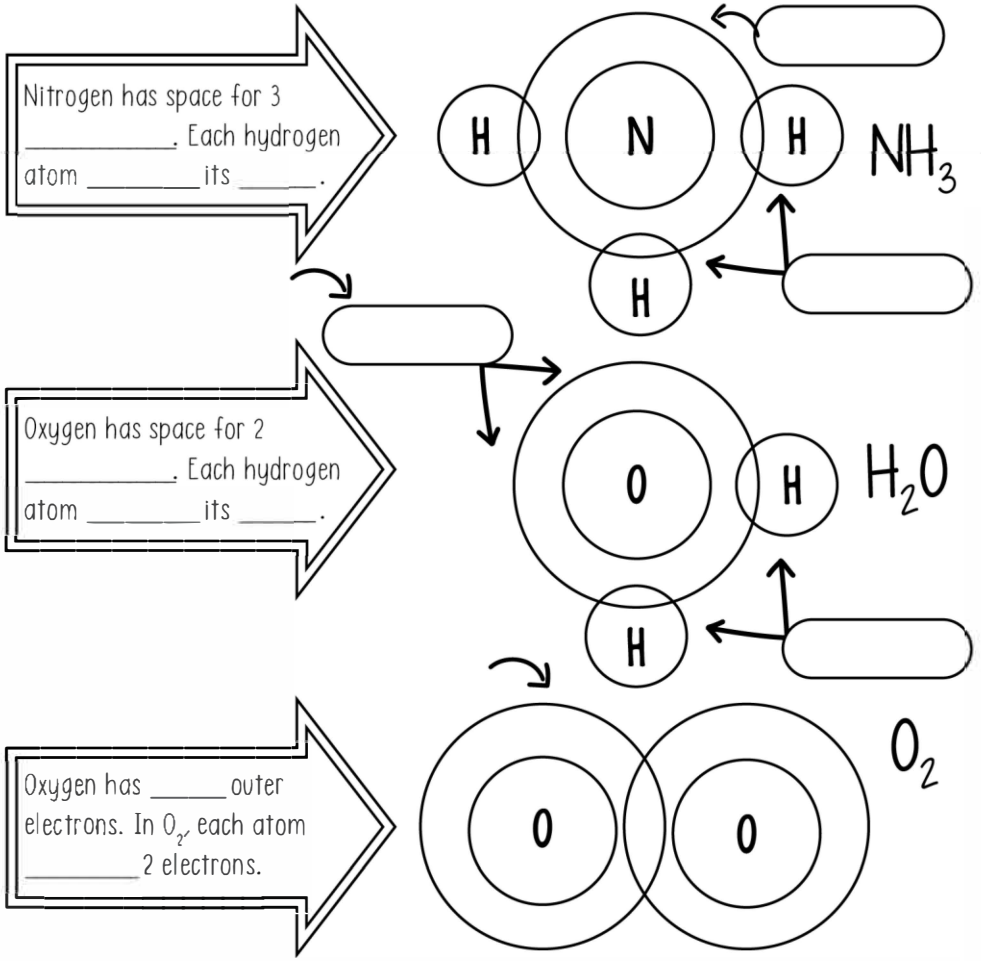
IONIC

Ionic bonding occurs when a _____ atom donates one or more _____ to a _____ atom. The metal becomes more _____ (_____) and the non-metal becomes more _____ (_____).

WHAT IS AN ION?

Atoms are normally _____ neutral, same number of _____ & _____. An _____ is an atom that has _____ or _____ an electron.

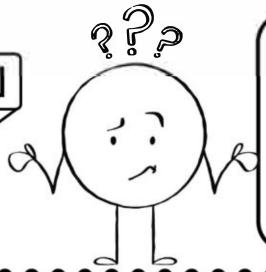




Covalent bonding occurs when atoms _____ pairs of _____ instead of giving and receiving. The _____ is included in the _____ shell of _____ atoms. Covalent bonding occurs between atoms of _____.

COVALENT

CHEMICAL BONDING

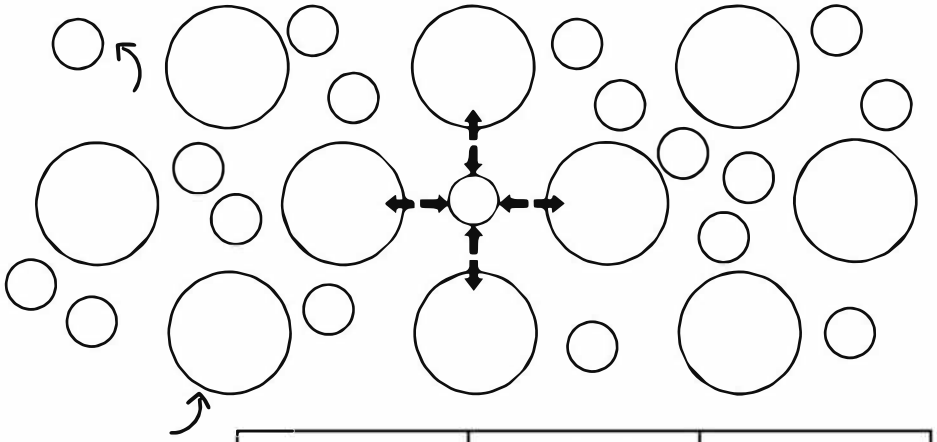


WHAT IS THE DIFFERENCE BETWEEN A COMPOUND AND A MOLECULE?

A compound is two or more _____ elements bonded together. A _____ is two or more _____ bonded together (they don't have to be _____) O_2 is a molecule but not a compound.

METALLIC

Metallic bonds occur when atoms of _____ give up _____ electrons, forming an electron "_____." The _____ charged atoms are _____ through their _____ to the _____ charged electrons.



	IONIC	COVALENT	METALLIC
Electrons			
Bond			
State			
Conductivity			
Melting Point			
Examples			

You try: PRACTICE

What is the most important factor affecting how atoms form chemical bonds? Why?

Atoms of which elements tend to gain electrons? Atoms of which elements tend to lose electrons?

When a Cl atoms gains an electron, it gets a charge of _____ and is known as a(n) _____

What is the Octet Rule?

Identify each of the following as ionic (I) or covalent (C).

C_4H_{10}	Al_2O_3	Na_2S	H_2S
_____	_____	_____	_____
BaF_2	NO_2	CBr_4	$MgCl_2$
_____	_____	_____	_____

Homework

ASSIGNMENT #5: Reading about Compounds pg 22-23 Comparing Ionic + Covalent Compounds pg 24 & "Compounds Review" pg 25

This assignment is to be completed below in the space provided.



READING ABOUT COMPOUNDS

Complete the following reading about types of compounds and their properties. Be sure to "Mark the Text" and highlight KEY DEFINITIONS as you read along. ALSO, answer the "Reading Check" questions in the side margin as you go!

Mark the Text

Identify Definitions

Highlight the definition of each word that appears in bold type.

How do compounds form?

Recall that an element is a pure substance that is made up of one type of atom. A **compound** is a pure substance that is made up of two or more types of atoms that are joined together due to a chemical change. Water, sugar, and table salt are three examples of compounds.

Atoms are held together in compounds by chemical bonds. These chemical bonds are created by attractive forces between atoms. Chemical bonds are formed when atoms gain or lose electrons, or when they share electrons. Recall that an atom is electrically neutral. When an atom loses electrons it becomes positively charged. When an atom gains electrons it becomes negatively charged.

Reading Check

1. When is an ionic compound formed?

_____

_____

_____

_____

_____

What are ionic compounds?

If atoms gain electrons from other atoms or lose electrons to other atoms, they form **ionic compounds**. Ionic compounds usually form between metals and non-metals. Why? The atoms in metals tend to lose electrons. So metals have a positive charge when they form ions. The atoms in non-metals tend to gain electrons. So non-metals have a negative charge when they form ions.

Reading Check

2. Draw a Bohr Diagram of an Na^+ ion and a Cl^- ion. Circle the electron that Sodium loses and draw and arrow to show how that electron is given to chlorine.

How do ionic compounds form?

When atoms of a metal come near atoms of a non-metal, they may join together to form an **ionic compound.** Electrons from the metal atoms are transferred to the non-metal atoms to create oppositely charged ions that attract each other. For instance, think about what happens when a sodium atom (metal) comes near a chlorine atom (non-metal). The sodium atom loses an electron to form a positive ion, and the chlorine atom gains an electron to form a negative ion. The two oppositely-charged ions are attracted to each other.

Ionic compounds are made up of charged particles (ions), but the positive charges and the negative charges balance, so ionic compounds are neutral.

A repeating pattern of positive and negative ions in a compound is called an **ionic lattice**, sometimes also called a crystalline lattice.

How do covalent compounds form?

Sometimes atoms share electrons instead
atoms share electrons, they form covalen
compounds form when non-metal atoms bond together by sharing their .
electrons. Since the electrons are shared, the particles that make up covalent compounds are neutral. They do not have a charge. A neutral particle that is made up of atoms that are joined together by covalent bonds is called a **molecule**. A water molecule is a covalent compound. Its molecules are made of hydrogen and oxygen. Carbon dioxide gas is also a covalent compound. Its molecules are made of carbon and oxygen. ✔

✔ *Reading Check*

3. When is a covalent compound formed?

_____

_____

_____

_____

_____

What is a polyatomic ion?

Some ions contain more than one atom. For example, the nitrate ion (NO_3^-) contains nitrogen and oxygen. The carbonate ion (CO_3^{2-}) contains carbon and oxygen. In these many-atom ions, the atoms are held together with covalent bonds. But the many-atom unit has a charge, so it is considered an ion. An ion that is made up of two or more atoms that are held together with covalent bonds is called a **polyatomic ion**.



Use the terms in the READING to fill in the blanks.

1. A pure substance that is made up of one type of atom is called a(n) _____.

2. A pure substance that is made up of two or more types of atoms that are joined together due to a chemical change is called a(n) _____.

3. Atoms in a molecule and ions in an ionic lattice are held together by _____.

4. Chemical bonds are formed when atoms gain or lose _____ or when they share_____.

5. When an atom loses electrons it becomes _____ charged. When an atom gains electrons it becomes _____ charged.

6. Metals and non-metals may form _____.

7. The atoms in non-metals tend to _____ electrons.

8. A(n) _____ is a repeating pattern of positive and negative ions.

9. _____ form when non-metal atoms bond together by sharing their electrons.

10. A neutral particle that is made up of atoms that are joined together by covalent bonds is called a(n) _____.

11. A(n) _____ is an ion that is made up of two or more atoms that are held together with covalent bonds.

Comparing ionic and covalent compounds

Use the chart to help you compare ionic compounds and covalent compounds. On the left side, place the letters of the statements that are only true of ionic compounds. On the right side, place the letters of the statements that are only true of covalent compounds. In the middle, place the letters of the statements that are true of both compounds.

A. atoms gain or lose electrons to form ions

B. pure substance made up of two or more kinds of elements

C. compound is made of a positive ion and a negative ion

D. atoms join by sharing electrons

E. atoms are joined to each other by chemical bonds

F. exist as a solid in the form of an ionic lattice

G. oppositely charged ions attract each other

H. molecule made of uncharged atoms

I. bond between atoms is due to electron transfer

J. compound is made of a non-metal and a non-metal

K. sodium chloride (NaCl) is an example

L. water (H_2O) is an example

Ionic compound	Both	Covalent compound

Compounds Review

Match each Term on the left with the best Descriptor on the right. Each Descriptor may be used only once.

Term	Descriptor
1. _____ molecule	**A.** pure substance made of one type of atom
2. _____ ionic lattice	**B.** atoms combine by gaining or losing electrons
3. _____ polyatomic ion	**C.** repeating pattern of positive and negative ions
4. _____ ionic compound	**D.** atoms combine by sharing electrons to form molecules
5. _____ covalent compound	**E.** neutral particle that is made up of atoms that are joined together by covalent bonds
	F. ion made up of two or more atoms that are held together with covalent bonds

Circle the letter of the best answer.

6. Atoms in non-metals tend to gain

A. molecules

B. ions

C. atoms

D. electrons

7. Which of the following can be formed when there is electron transfer between metals and non-metals?

A. molecule

B. element

C. ionic bond

D. covalent bond

8. Which of the following is formed due to the sharing of electrons between two non-metals?

I.	a molecule
II.	a covalent bond
III.	a covalent compound

A. I and II only

B. I and III only

C. II and III only

D. I, II, and III

9. Water is a(n)

A. element

B. polyatomic ion

C. ionic compound

D. covalent compound

10. Sodium chloride is a(n)

A. element

B. polyatomic ion

C. ionic compound

D. covalent compound

11. Which of the following can be formed when a non-metal atom reacts with a non-metal atom?

A. element

B. polyatomic ion

C. ionic compound

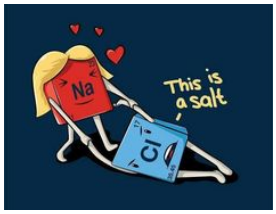
D. covalent compound

PART I: WHAT ARE IONIC COMPOUNDS?

Atoms want to _____ to have a full outer shell (_____). A full outer shell is a _____ electron arrangement. When this happens atoms become _____.

When a _____ gives its electron/s to a _____ they join to form an _____.

The ion charge is the number of electrons an atom must _____ or _____ to have a stable electron arrangement (full _____ shell).



ALL atoms want a _____

Example: sodium and chlorine → sodium chloride (NaCl)

- Na has _____ electron in its valence shell and Cl has _____.
- Sodium needs to lose _____ electron and chlorine needs to gain _____ electron.
- The electron is transferred, and two ions form both having _____ outer orbitals.
- The ions are charged: _____ has a _____ charge, and _____ has a _____ charge.



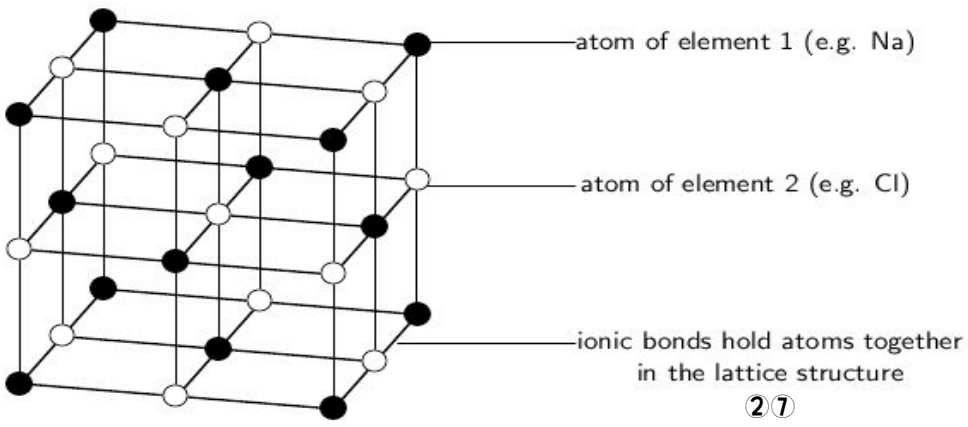
- Atoms with _____ ion charges (they lose electrons) are the _____. They form positive _____.
- Atoms with negative ion charges (they _____ electrons) are the non-metals. They form negative _____.
- Sometimes atoms need to lose or gain ***more than one*** _____.
    - For example, magnesium needs to lose _____ electrons and oxygen needs to gain _____ electrons. The solution is to use one of each atom but to transfer _____ electrons!

Example: magnesium and oxygen → magnesium oxide (MgO)

- But what happens when the electron requirements are different? For example, magnesium needs to lose _____ electrons whereas chlorine only needs to gain _____ electron. The solution is to use _____ chlorine atoms for every _____ magnesium atom.

Example: magnesium and chlorine → magnesium chloride ($MgCl_2$)

Structure of Ionic Compounds



PRACTICE Study the images below and answer the question at the end of the caption.

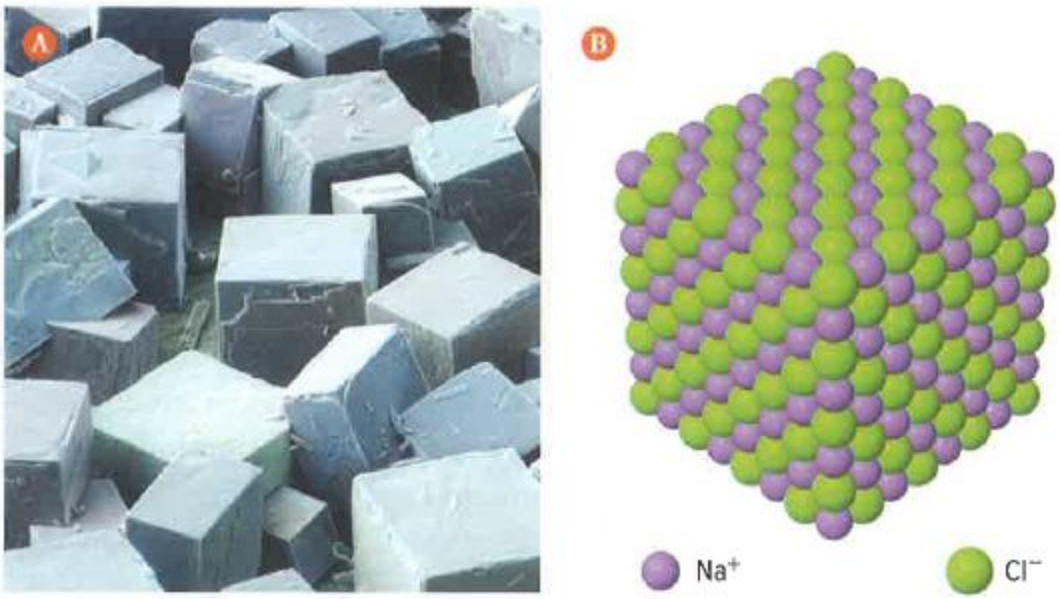


Figure 2.24 A This image shows the cubic structure of sodium chloride crystals. Each crystal contains millions and millions of sodium ions and chloride ions. B Sodium chloride crystals consist of sodium and chloride ions arranged in a repeating pattern. **Sodium chloride is made of charged particles, but the compound overall has no charge. Why?**

PART J: PROPERTIES OF IONIC COMPOUNDS

1. **Ionic compounds have** _____.

   Melting an ionic compound requires overcoming strong _____

   _____.

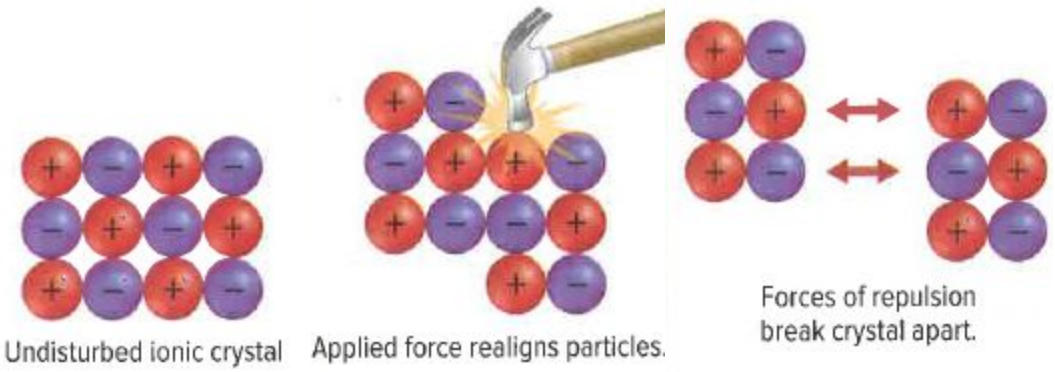
   Because these bonds are strong, a great deal of energy is required to break them. As a result, ionic compounds tend to melt only at very _____.

   For example the melting point of sodium chloride (NaCl) is _____.

2. **Ionic compounds are** _____.

Because of the strength of ionic bonds, ionic solids are very hard. But when enough force is applied the ions will shift out of alignment. This causes ions with the same charge to be close together. These ions will then _____.

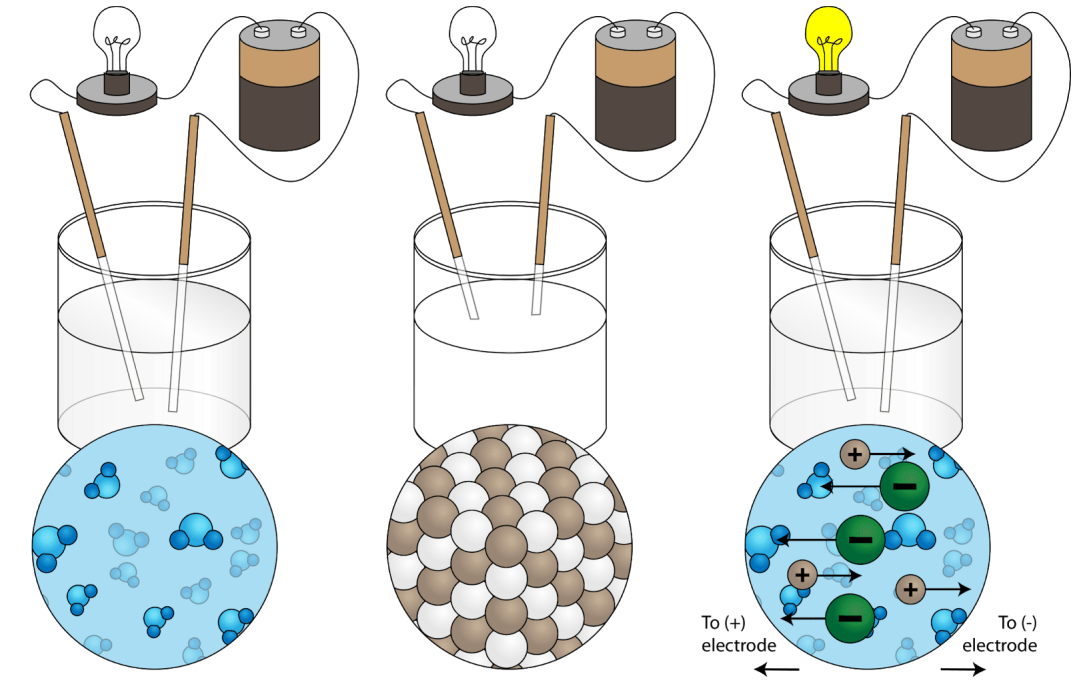
Figure 2.25 When a force strong enough to overcome the strong forces of attraction between oppositely charged ions is applied, ions with like charges come close together. They repel one another and the solid cracks.



3. **Ionic compounds conduct** _____.

In these states, the ions that make up an ionic compound are free to _____

and can therefore conduct electricity.



A Distilled water does not conduct a current.

B Positive and negative ions fixed in a solid do not conduct a current.

C In solution, positive and negative ions move and conduct a current.



ASSIGNMENT #6: Ionic Compounds Summary pg 30

This assignment is to be completed below in the space provided.

Summary:

1. **How do you know if something is an ionic compound?**

2. **What is happening with the electrons in the formation of an ionic compound?**

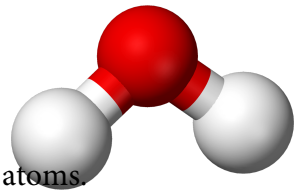
3. **What are some general properties of ionic compounds?**

Draw the Bohr diagrams of the following ionic compounds below. Start with the atoms of each, then draw the ions.

lithium fluoride (LiF)

beryllium sulphide (BeS)

PART K: WHAT ARE COVALENT COMPOUNDS?



The atoms of many _____ **share electrons** with other non-metal atoms.

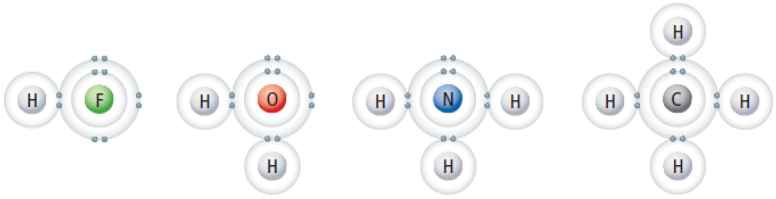
In covalent bonding, atoms overlap slightly, and one u_____electron from each atom will pair together.

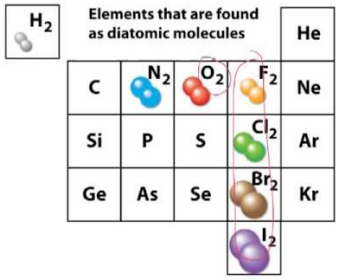
Both atoms are attracted to the same ***pair of electrons,*** forming a c_____ bond. A covalent c_____is formed when non-metallic atoms ***share electrons*** to form covalent bonds.

A covalent m_____ is a ***group of atoms*** in which the atoms are bound together by sharing one or more pairs of electrons.

The pair of electrons involved in a covalent bond are sometimes called the b_____ pair. A pair of electrons in the v_____shell that is ***not used in bonding*** is sometimes called a l_____ pair *(unpaired).*

Bohr Diagram:





DIATOMIC MOLECULES

A _____ molecule is a pair of atoms that are joined by _____bonds.

Diatomic elements form this way because the two-atom molecules are m_____s_____ than the individual atoms.

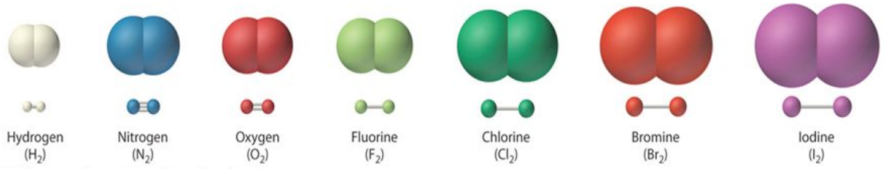
For example, fluorine gas is diatomic.

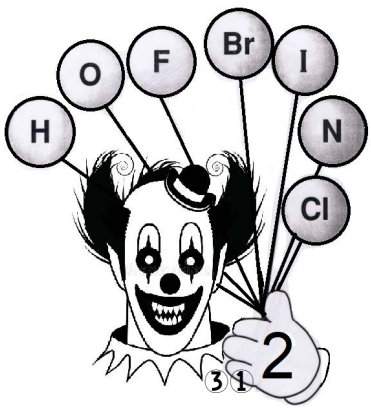
By joining together to form _____, each fluorine atom can achieve a full v_____ shell of _____ electrons

Other diatomic elements are hydrogen (H_2), nitrogen(N_2), oxygen (O_2), chlorine (Cl_2), bromine (Br_2), and iodine (I_2).

The HOFBrINCl Elements:

- There are 7 naturally occurring elements that , when found in nature, exist as _____ molecules.
- The atoms will not exist alone, they **bond** to each other.
- This means they must ***always*** be written with a subscript of _____.





PART L: PROPERTIES OF COVALENT COMPOUNDS

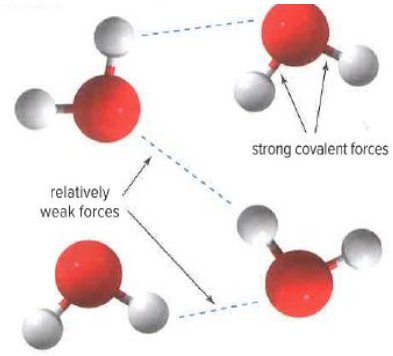
Covalent compounds have widely varying properties.

The plastic casing of a ballpoint pen, the components of gasoline, the strongly scented components in a banana, and the carbon dioxide that we exhale with every breath are all covalent compounds.

Here are some properties that covalent compounds do share:

1. **Covalent compounds have** _____.

Although the forces that hold atoms together in molecules are very strong, the bonds that attract one molecule to another in a covalent compound are relatively _____.



Therefore, when you _____ or _____ a covalent compound there is less

of a barrier to overcome so they will melt _____.

2. **Covalent compounds are relatively** _____.

The _____ of the forces between molecules also explains the relative softness of covalent compounds.

3. **Covalent compounds are poor** _____.

Unlike ionic compounds, covalent compounds do not have free _____ or ions. This covalent compounds poor conductors of both heat and electricity.

Properties of Ionic Compounds	Properties of Covalent Compounds



LAB: COMPARING PROPERTIES OF IONIC AND COVALENT COMPOUNDS

Introduction

Chemical compounds are combinations of atoms held together by chemical bonds. We have studied two basic types of compounds: ionic and covalent. Ionic compounds result when one or more electrons from one atom or group of atoms are transferred to another atom or group of atoms. In covalent compounds the electrons are shared by the bonded atoms.

The curls and waves of your hair are the result of many ionic, covalent and hydrogen bonds between the chains of atoms that make up each protein filament. Styling hair by wetting it or heating it with a curling iron is an attempt to change the hydrogen and ionic bonds. The solution in a “permanent” (or “perm”) on the other hand, breaks and reforms covalent bonds. A permanent wave does not wash out when you shampoo your hair.

The physical properties of a substance such as melting point and solubility tell us a lot about the type of bond in a compound. In this experiment, you will conduct tests on physical properties and compile data to enable you to classify compounds as ionic or covalent. You will also be given a unknown compound to test and predict whether it is an ionic or covalent compound.

Objective:

Materials

A - sodium chloride	aluminum plates x 5
B - sucrose ($C_{12}H_{22}O_{11}$)	hot plate
C - potassium chloride	5 test tubes
D - acetaminophen ($C_8H_9NO_2$)	tongs
E - unknown	sharpie
water	

Procedure

Part 1: Melting Point

1. Put on your safety glasses and keep them on until you have finished cleaning up your lab station.
2. Label your aluminum trays A, B, C, D, E.
3. Place a very **SMALL** sample of each substance A, B, C, D, E on the matching aluminum tray.
4. Plug in your hot plate and turn it on medium/high.
5. Place two of the aluminum trays on the hot plate. Observe the substances in each tray and make note of which substances melt.
6. If the substance does not melt right away continue to observe for 4 minutes.
7. Using the tongs, remove aluminum trays from the hot plate and continue the procedure with your next two samples.

Part 2: Solubility

1. Place one **SMALL** scoop of each substance A, B, C, D, E into a separate clean test tube.
2. Fill the test tube ⅓ full with water and gently agitate the test tube for 3 minutes.
3. Record your observations. Repeat this procedure for each of the test tubes.

Data

Substance	**Colour**	**Melting Point: Does it melt?**	**Solubility: Does it dissolve?**	**Ionic or Covalent**
A - sodium chloride **chemical formula: _____**				
B - sucrose ($C_{12}H_{22}O_{11}$)				
C - potassium chloride **chemical formula: _____**				
D - acetaminophen ($C_8H_9NO_2$)				
E - unknown				

Analysis

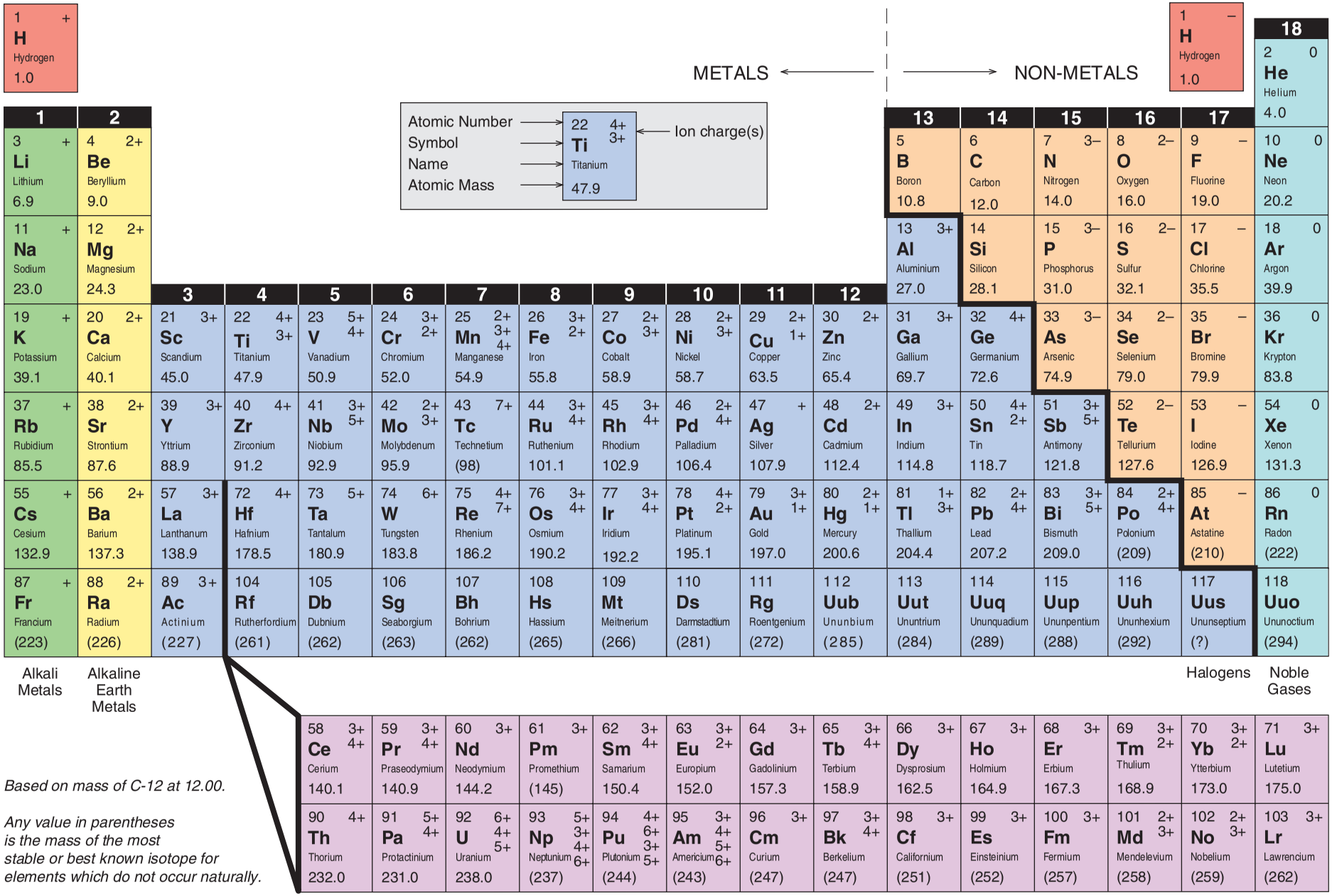
1. Look carefully at your results. Do you see any patterns? Summarize the results from your experimental data on the last page based on whether you were observing an ionic or covalent compound. If you don't see any patterns, state "no pattern". If you do see general patterns, state them.

	Ionic	**Covalent**
Melting Point		
Solubility		

2. Based on your experimental data and your analysis above, do you think the unknown compound was ionic or covalent? Why?

3. Can you think of another property we could test to determine if an unknown substance is an ionic compound or a covalent compound? Have a look at your notes for both types of compounds if you can't think of anything.

PERIODIC TABLE OF THE ELEMENTS



Based on mass of C-12 at 12.00.

Any value in parentheses is the mass of the most stable or best known isotope for elements which do not occur naturally.

NAMES, FORMULAE AND CHARGES OF SOME POLYATOMIC IONS

Positive Ions	Negative Ions	
NH_4^+ Ammonium	CH_3COO^-	Acetate
	CO_3^{2-}	Carbonate
	ClO_3^-	Chlorate
	ClO_2^-	Chlorite
	CrO_4^{2-}	Chromate
	CN^-	Cyanide
	$Cr_2O_7^{2-}$	Dichromate
	HCO_3^-	Hydrogen carbonate, bicarbonate
	HSO_4^-	Hydrogen sulfate, bisulfate
	HS^-	Hydrogen sulfide, bisulfide
	HSO_3^-	Hydrogen sulfite, bisulfite
	OH^-	Hydroxide
	ClO^-	Hypochlorite
	NO_3^-	Nitrate
	NO_2^-	Nitrite
	ClO_4^-	Perchlorate
	MnO_4^-	Permanganate
	PO_4^{3-}	Phosphate
	PO_3^{3-}	Phosphite
	SO_4^{2-}	Sulfate
	SO_3^{2-}	Sulfite

NAMES AND FORMULAE OF COMMON ACIDS

Hydrochloric acid	HCl
Sulfuric acid	H_2SO_4
Nitric acid	HNO_3
Acetic acid	HCH_3COO

PREFIXES

1	mono
2	di
3	tri
4	tetra
5	penta
6	hexa
7	hepta
8	octa
9	nona
10	deca